

Minimization of Dead-Time Effect in Current-Controlled Three-Phase PWM Inverters by Using Virtual Inductor

Adinda Ihsani Putri, Arwindra Rizqiawan, Tri Desmana Rachmildha, Yanuarsyah Haroen, and Pekik Argo Dahono

School of Electrical Engineering and Informatics
Institut Teknologi Bandung
Bandung, Indonesia
adindaip@students.itb.ac.id¹

Abstract: Dead-time causes distorted output current in current-controlled three-phase PWM inverters. Instead of using a real inductor, a virtual inductor is proposed in this paper to minimize the dead-time effects. Virtual inductor is an additional controller block that changes the system behavior so that it is similar as if there is a real inductor in the system. As it is virtual inductor, we have no problems of cost and losses. How to design a virtual inductor for dead-time effect minimization is presented in this paper. The effect of low-pass filter (LPF) which is mandatory in virtual inductor application is also presented in this paper. Simulation and experimental results are included to show the effectiveness of proposed virtual inductor.

Keywords: Dead-time effect; current-controller; inverter; virtual inductor.

1. Introduction

Current controller plays an important role in inverter applications such as high-performance ac drives and grid-connected inverters. In grid-connected applications, the current controller is designed so that the THD of output current does not exceed the limit that is allowed by grid code [1], [2]. The current controller must also track the current reference tightly to follow active and reactive currents needed by the grid [3], [4]. In motor control, high performance current controller affects torque response and robustness [5]–[7].

There are two types of current controller, linear and non-linear [8]. Linear current controller is mostly used since it allows the use of modulators with constant frequency such as carrier-based PWM. Hysteresis and predictive current controllers are non-linear. Hysteresis current controller is easy to be implemented but it suffers undefined switching frequencies [9]–[11]. Predictive current controller can track the reference tightly and has a good dynamic performance, but it is complex and highly requiring accurate model [1], [6], [12], [13]. In this paper, a linear synchronous PI current controller is used because of its capability to produce a zero steady-state error. However, the nonidealities of inverter, such as dead-time, can inhibit the PI current controller from working properly.

Dead-time delays the ON signals for the inverter switching devices. During the dead period, both upper and lower switching devices in an inverter leg are turned OFF in order to prevent a short-circuit through an inverter leg. Though it is safe for the inverter, dead-time raises a new problem to the inverter in the form of harmonics and instabilities [14]. The dead-time results in low order harmonics at the output voltage and current. These low order harmonics are getting worse when the inverter works at low modulation indexes. Intuitively, an inductor can be used to overcome this low-order harmonic problem. However, to filter the low order harmonics, the inductor must be large, which makes the system bulky, costly, and lossy.

There are several methods that have been proposed to overcome dead-time effect without inductor. Generally, the methods can be categorized into two, feedforward and feedback. In feedforward method [15], [16], reference signal is modified before being forwarded to the PWM generator. However, this method requires accurate mathematical model to compensate the dead-time, meanwhile the dead-time effect changes with the operating condition. In

feedback method [13], [17], an additional voltage or current sensor is needed. Since there is existing current sensor in the system, there is no need for additional sensor if current feedback is used. The drawbacks of this system are more complex and cannot be used as a plug-in controller. Indeed, PI controller can be designed to suppress low frequency disturbance by increasing proportional gain. But the method can increase sensitivity and disrupt stability of the system.

This paper proposed a new way to minimize dead-time effect by using virtual inductor. Virtual inductor is an additional controller that can be plugged-in to the existing system so that the system mimics the system behavior as if there is a real inductor connected to the system circuit [18]. Since the virtual inductor is just an additional controller block in current controller, there is no problem with losses and cost. Furthermore, virtual inductor can be designed in such a way to suppress only the dead-time effect without affecting the current controller performance. The effects of LPF on the system performance is also analyzed in this paper. The analysis includes system stability, sensitivity and steady state error. The effectiveness of virtual inductor in minimizing the dead-time effects are verified through simulation and experiment.

2. Dead-time Effect in Current-Controlled Three-Phase PWM Inverter

Figure 1 shows the current-controlled three-phase PWM inverter system. It can be seen that we have two current controllers work separately in direct and quadrature reference frames. The current controllers work in synchronous reference frame and, therefore, works with dc quantities. Under ideal conditions, the steady-state errors of the current are zero.

Figure 2 shows simulated results of the output currents without and with dead-time. Without dead-time, the output currents are almost sinusoidal with high switching-frequency ripple. In dq frame, the dq currents are almost pure dc currents. With dead-time, the output currents are distorted by low-order harmonics. These low-order harmonics cannot be reduced by increasing the switching frequency. Though the low-order harmonics can be reduced by using an output filter inductor, the required inductor will be very large.

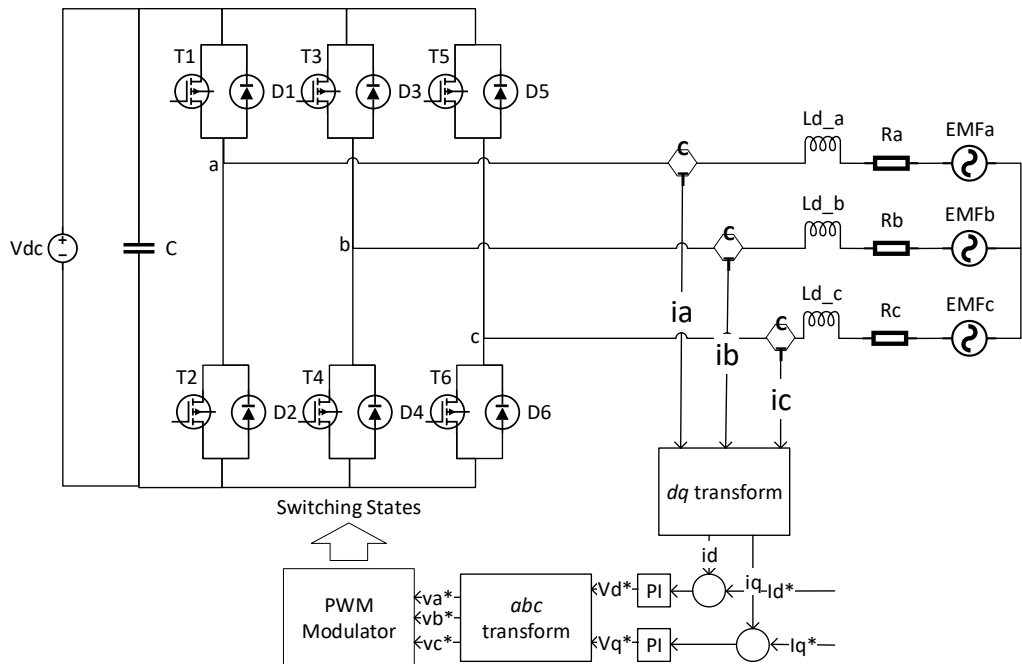


Figure 1. Current-controlled three-phase PWM inverter.

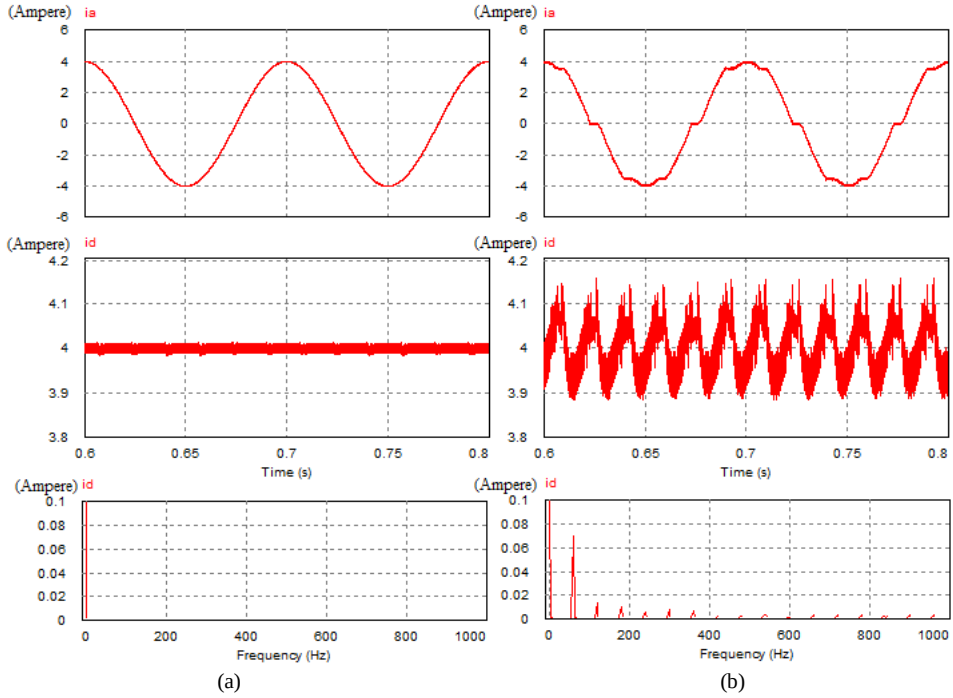


Figure 2. Output current; (a) without dead-time and (b) with dead-time.

3. Virtual Inductor Concept

Block diagram in figure 3 represents a current-controlled three-phase PWM inverter. There are PI current controller, coupling between d -axis and q -axis, and the plant that is represented by L_d and R . The grid voltage or motor emf can be taken into account in the disturbance signal N_s . To develop the proposed virtual inductor, the block diagram in figure 3 is simplified by assuming R , emf and coupling between d -axis and q -axis as disturbance signal N_s . This disturbance signal can also be used to represent the dead-time effects. An inductor L_v is added in series to L_d with the aim of minimizing the distortion of output current caused by dead-time effect. The plant is then represented by L_d and L_v . The result of this simplification is shown in figure 4(a). Later, the development of virtual inductor will be explained in d -axis.

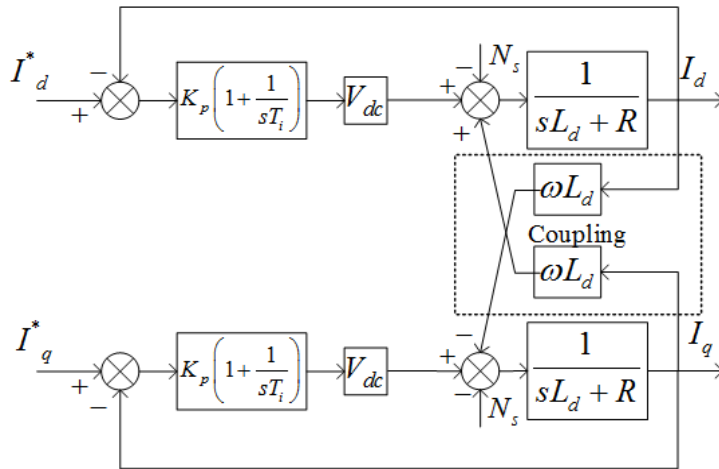


Figure 3. Block diagram of current-controlled three-phase inverter

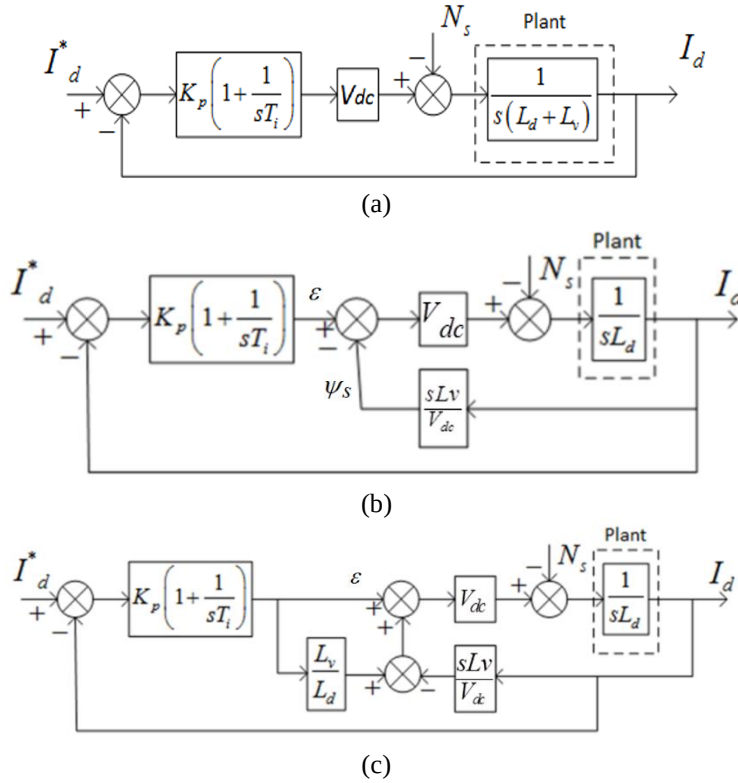


Figure 4. Development of virtual inductor.

Block diagram in figure 4(a) is then modified in such a way so that L_v is not the part of the plant but the control block. In this case, the modification does not change the overall system response and the result is shown in figure 4(b). System response of block diagram in figures 4(a) and (b) is defined in (1)

$$I_d = \frac{\frac{sV_{dc}K_p}{(L_d + L_v)} + \frac{V_{dc}K_p}{(L_d + L_v)\tau_i}}{\left(s^2 + s\frac{V_{dc}K_p}{(L_d + L_v)} + \frac{V_{dc}K_p}{(L_d + L_v)\tau_i}\right)} I_d^* - \frac{\frac{s}{(L_d + L_v)}}{\left(s^2 + s\frac{V_{dc}K_p}{(L_d + L_v)} + \frac{V_{dc}K_p}{(L_d + L_v)\tau_i}\right)} N_s \quad (1)$$

From (1), it can be seen, that the effect of L_v is found both in current reference and dead-time effect. If it is assumed that ψ_s in figure 4(b) is the output of virtual inductor block that affects the current reference and ε is the output of PI current controller, then ψ_s can be defined as in (2).

$$\psi_s = \frac{L_v}{L_d} \varepsilon - \frac{L_v}{L_d V_{dc}} N_s \quad (2)$$

To cancel the effect of L_v to the reference current, L_v/L_d block is added to be multiplied by the output of PI current controller. The result is added to the output of PI current controller as shown in figure 4(c). The system response of block diagram in figure 4(c) is defined in (3).

$$I_d = \frac{\frac{sV_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i}}{s^2 + s\frac{V_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i}} I_d^* - \frac{\frac{s}{(L_d + L_v)}}{\left(s^2 + s\frac{V_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i}\right)} N_s \quad (3)$$

L_v component is no longer affecting the current reference. It can be shown easily that the virtual inductor in (3) is the same as a feedback compensation that is derived by using a disturbance observer concept if the virtual inductor L_v is equal to the actual inductor L_d . Thus, disturbance observer is a special case of virtual inductor concept. Figure 5. 5(a) shows the system responses to current reference (I_d^*) are similar at various values of L_v . Figure (b) shows the system response to dead-time effect (N_s). It can be seen, that the dead-time effect is effectively suppressed when the virtual inductor L_v is larger. This verify that virtual inductor has been successfully designed to minimize the dead-time effect without affecting the reference current. In practice, we use a differentiator to implement the virtual inductor and, therefore, the virtual inductance cannot be too large. To suppress the effects of noise, we need a low-pass filter (LPF) to implement the virtual inductor. The effect of LPF in the implementation of virtual inductor will be discussed in the next section.

4. The Effect of Low Pass Filter

In order to reduce the noise effects, a low pass filter (LPF) is added to the input of virtual inductor as seen in figure 6. A second order LPF is used in this paper and the transfer function is shown in (4).

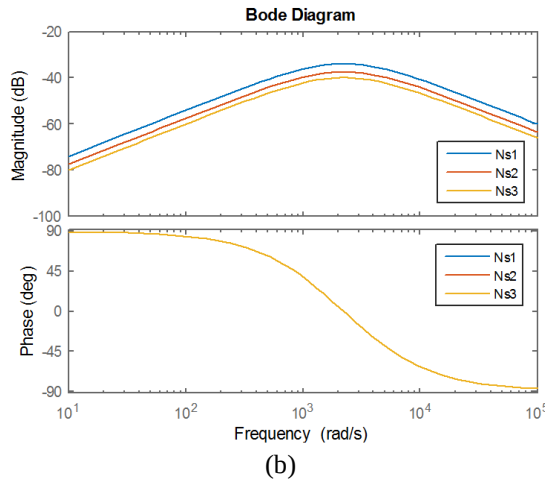
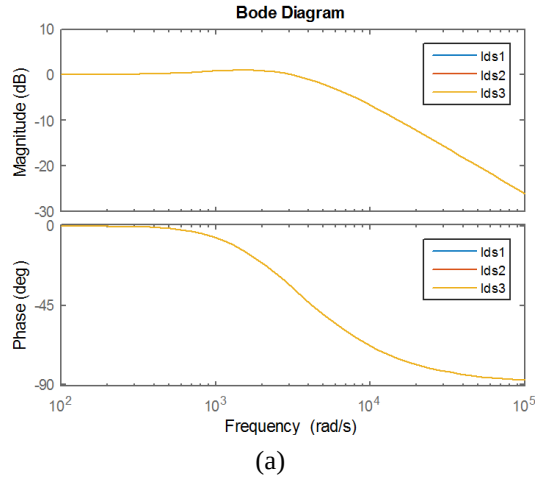


Figure 5. Bode plot of current-controlled three phase system with virtual inductor; (a) reference current (I_d^*) and (b) dead-time effect (N_s)

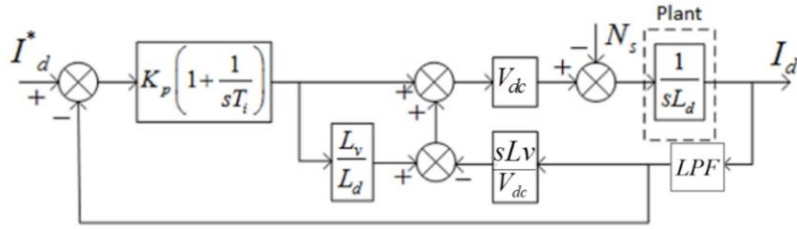


Figure 6. Block diagram of current controlled three-phase inverter with virtual inductor and LPF.

$$LPF = \frac{k\omega_c^2}{s^2 + 2\zeta\omega_c s + \omega_c^2} \quad (4)$$

where k is gain, ω_c is cut-off frequency and ζ is damping ratio. The transfer function of the system with LPF is shown in (5).

$$I_d = \frac{\frac{sV_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i}}{LPF \left[\frac{\left(\frac{L_d}{LPF} + L_v \right)}{(L_d + L_v)} s^2 + s \frac{V_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i} \right]} I_d^* \quad (5)$$

$$- \frac{s}{(L_d + L_v)LPF \left[\frac{\left(\frac{L_d}{LPF} + L_v \right)}{(L_d + L_v)} s^2 + s \frac{V_{dc}K_p}{L_d} + \frac{V_{dc}K_p}{L_d\tau_i} \right]} N_s$$

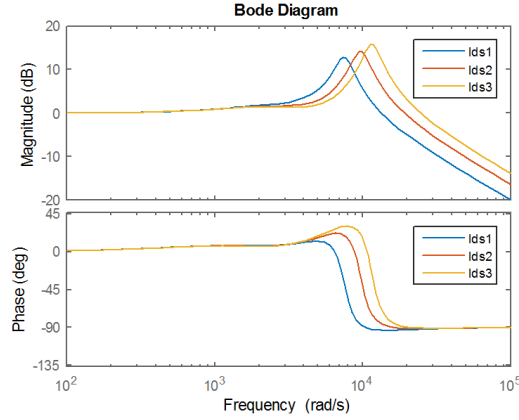
If the transfer function of LPF is unity, the transfer function in (5) is the same as in (3).

To verify the effects of LPF to the system, bode plot of the system with and without LPF are presented in figure 7. The parameters are $V_{dc} = 100$ V, $K_p = 0.25$, $T_i = 0.001$, $L_d = 5$ mH, $k = 1$, $\zeta = 0.707$ and $\omega = 2\pi 1000$ rad/s. Different values of L_v are used to determine the effect of virtual inductor, whereas $L_{d1} = L_d$, $L_{d2} = 2L_d$, and $L_{d3} = 3L_d$. At lower frequency, there is no significant effect of different values of L_v to reference current. However, at higher frequency, the input will be amplified so that it can cause larger current error. The similar response is found in system response to disturbance. At lower frequency, larger L_v suppress disturbance which includes dead-time effect better. At higher frequency, larger L_v amplifies other disturbances such as switching frequency and noises. In addition to afore-mentioned system responses, it is necessary to consider the effect of virtual inductor to system stability, sensitivity and steady-state error so that the proper value of L_v can be selected.

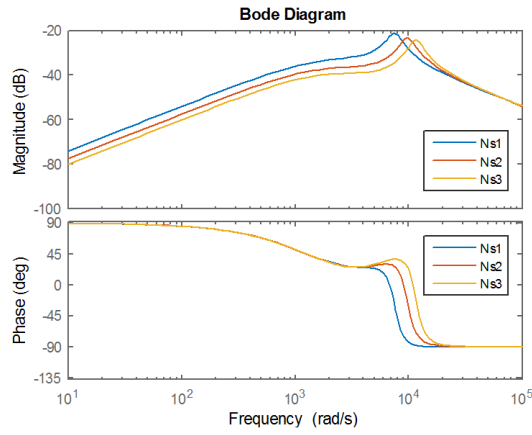
A. Stability Analysis

The Nyquist stability criterion is used for stability analysis. Stability of a closed-loop system is analyzed through Nyquist plot of its open-loop system. Figure 8 is modified version of system in figure 6. This system can be seen as if there are two closed-loop systems, the inner one and the outer one. To get the open loop transfer function, the closed loop transfer function of the inner one must be obtained first and then the open loop system obtained by using (6). The result is written on (7)

$$G(s)H(s) = [G_1(s)G_{in}(s)]H_1(s) \quad (6)$$



(a)



(b)

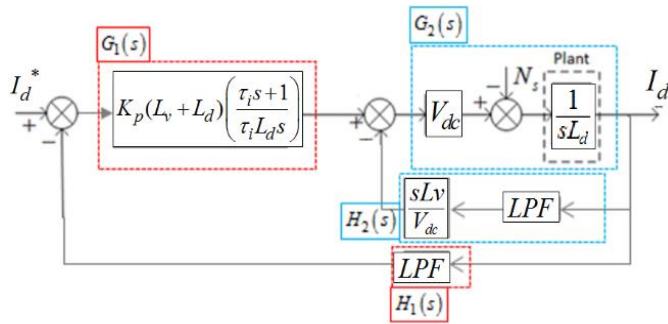
 Figure 7. Bode plot of current-controlled three phase system with virtual inductor with LPF effect; (a) reference current (I_d^*) and (b) dead-time effect (N_s).


Figure 8. Closed-loop system of current-controlled three-phase PWM inverter.

$$G(s)H(s) = \frac{\frac{K_p E_d}{L_d} s + \frac{K_p E_d}{\tau_i L_d}}{\left(\frac{L_d}{LPF} + L_v \right) s^2} \quad (7)$$

From the open loop equation, at any L_v values, the poles are always zero. So it can be concluded that at different values of L_v , the system is always stable. Hence L_v doesn't affect stability.

B. Sensitivity Analysis

Dead-time is categorized as disturbance and potentially causes the system unstable. The sensitivity analysis is required to measure how robust a system in responding the dead-time effect. A system is called too sensitive if there is a small change in input gain or phase that causes the system to be unstable. For the proposed system, the sensitivity function is obtained by using sensitivity function which is shown in (8) and the result is shown in (9).

$$S = \frac{1}{1+G(s)H(s)} \quad (8)$$

$$S = \frac{s^2}{s^2 + \frac{K_p E_d}{L_d} \left(\frac{L_d + L_v}{L_{PF}} + L_v \right) s + \frac{K_p E_d}{\tau_i L_d} \left(\frac{L_d + L_v}{L_{PF}} + L_v \right)} \quad (9)$$

Sensitivity analysis is done by observing the bode diagram of sensitivity function that is shown in figure 9. It can be seen that maximum value of sensitivity decreases as virtual inductance increases. On the other hand, this maximum value of sensitivity (S_{max}) is shifting to higher frequency, making the system more sensitive to higher frequency disturbance. It can be said that this system is not sensitive to dead-time effect which is low order frequency. However, this system is more sensitive to noise and switching frequency.

C. Steady-state Error Analysis

The steady-state error analysis is required to determine the ability of the system to follow certain types of inputs. There are several types of inputs; step, ramp parabolic, etc. In this paper the input is step. A steady-state error function in figure 8 is defined in (10).

$$E_{ss} = I_d^* - I_d L_{PF} \quad (10)$$

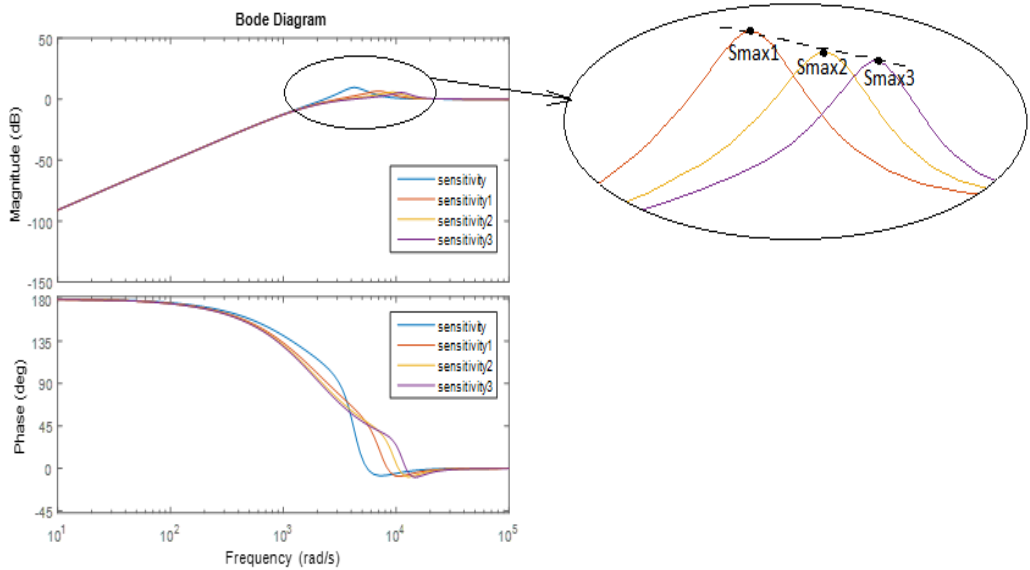


Figure 9. System sensitivity.

Inserting values of I_d^* and I_d , definition of steady state error is defined in (11).

$$E_{ss} = \frac{s^2}{s^2 + \frac{K_p E_d}{L_d} \left(\frac{L_d + L_v}{\frac{L_d}{L_{PF}} + L_v} \right) s + \frac{K_p E_d}{\tau_i L_d} \left(\frac{L_d + L_v}{\frac{L_d}{L_{PF}} + L_v} \right)} I_d^* \quad (11)$$

To calculate the value of steady-state error, s is assumed to be almost zero. The result is defined in (12).

$$\lim_{s \rightarrow 0} E_{ss} = \frac{1}{\frac{K_p E_d k}{L_d} \left(\frac{L_d + L_v}{L_d + L_v k} \right) \left(1 - \frac{1}{\tau_i \omega_c^2} \right)} \quad (12)$$

In this research, the value of k is 1, causes $(L_d + L_v)/(L_d + L_v k)$ is equal to 1. This causes L_v no longer affect the steady-state error of the system.

5. Experimental Results

A small experimental system was constructed to verify the proposed virtual inductor concept. Experimental parameters are shown in Table 1. In addition, the LPF that is used in this experiment is similar to (4). Whereas $k=1$, $\zeta=0.707$ and $\omega_c=1\text{kHz}$. Due to dead-time effect, it can be seen in figure 10(a) that the output current (i_a) is distorted with THD of 14.8%. Virtual inductor is then implemented to the system. The values of virtual inductance L_v are L_d , $2L_d$ and $3L_d$. The results can be seen in figure 10(b), (c) and (d). The measured d -axis output current spectrum is also shown in figure 11. The current harmonics are getting better when L_v is getting

Table 1. Experimental data.

Parameters	Value
DC Voltage (E_d)	100 Volt
Inductance (L_d)	5 mH
Referene current (I_d^*)	0.5 A
Fundamental frequency (f)	10 Hz
Switching frequency (f_s)	10 kHz
Proportional gain (K_p)	5
Integral time constant (τ_i)	0.001

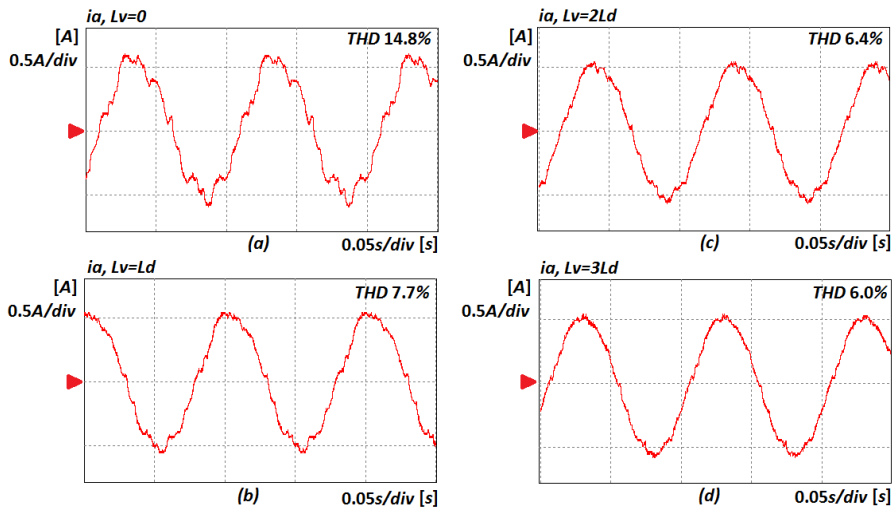


Figure 10. Measured output currents.

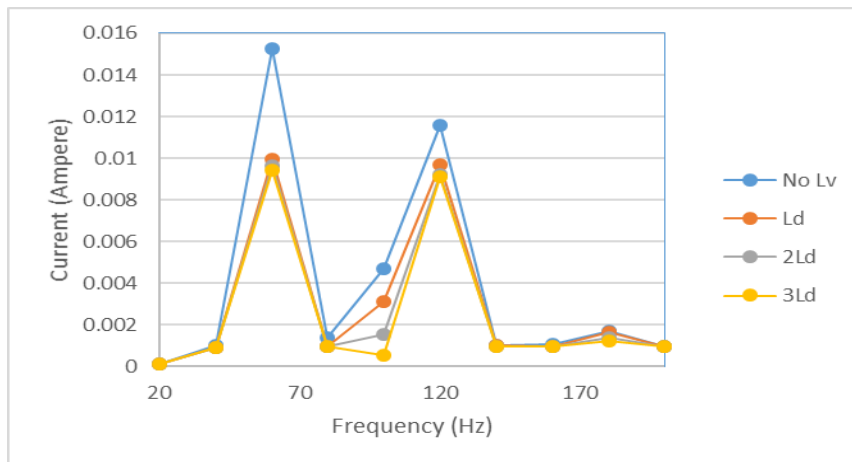


Figure 11. Spectrum of output current harmonics.

larger. It reduces from 14.8% at no L_v to 6.0% at L_v is equal to $3L_d$. The spectrum of harmonics are lower at low frequency when the virtual inductor is applied. The maximum value of L_v that is used in this experiment is no larger than $3L_d$, since it is the maximum value of L_v that causes better suppression of THD. If it is larger than $3L_d$, the THD is worse due to LPF effect. Larger L_v amplify the high frequency inputs, they are noises and switching frequency that contributes the harmonics to system. It also can be seen that in this experiment, the current is always follow the reference which shows that the system is always stable. There is also no change in average value of L_d , which shows that there is no steady-state error due to L_v .

6. Conclusion

Dead-time effect in current-controlled three-phase PWM inverter causes the low order harmonics in output current. Instead of using real inductor, a virtual inductor is proposed to minimize dead-time effect. Virtual inductor can be designed in such a way to only minimize the dead-time effect without affecting the current controller performance. The effects of low-pass filter on the virtual inductor performance is reported in this paper. The results show that virtual inductor affects only system sensitivity, not the stability and steady-state error. Experimental results have shown that the dead-time effects can be effectively suppressed by using the proposed virtual inductor when its value is no larger than $3L_d$. If it is larger, the THD is worse due to the contribution of amplified noises and switching frequency which is categorized as high frequency inputs for the system.

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Adinda Ihsani Putri, She received the bachelor degree (S.T.) from the Department of Power Engineering, Institut Teknologi Bandung, Indonesia, in 2010, and the M.Eng. degree from the School of Electrical Engineering, Chungbuk National University, Cheongju, South Korea, in 2012. She is currently a doctoral student in Institut Teknologi Bandung, Indonesia. She has been working in inverter research since 2010. The subjects are multiphase inverter, MPPT for Wind Power Generator, EV controller and minimization of non-idealities in inverter.



Arwindra Rizqiawan, He received his bachelor and master degree from Insitut Teknologi Bandung, Indonesia in 2006 and 2008, respectively, and doctoral degree from Shibaura Insitute of Technology, Japan, in 2012, all in the field of electrical engineering. His current main interests are power engineering, power electronics, and renewable energy.

He is currently serving as assistant professor in School of Electrical Engineering and Informatics, Institut Teknologi Bandung, Indonesia. He is certified professional engineer (IPM) in Indonesia by the Institution of

Engineers Indonesia (PII), and ASEAN Engineer by ASEAN Engineering Register.



Tri Desmana Rachmildha, He received B.Eng and M.Eng degrees in electrical engineering from Bandung Institute of Technology (ITB), Indonesia, in 1998 and 2002, respectively. He received Doctor Degree in Electrical Engineering from Joint PhD Supervision Program between Institut Nationale Polytechnique de Toulouse – Ecole Nationale Superieure d'Electrotechnique, d'Electronique, d'Informatique, d'Hydrolique, et de Telecommunication (INPT-ENSEEIH, France) and School of Electrical Engineering and Informatics, Bandung Institute of Technology, Indonesia, in

2009. He is a researcher at Electrical Energy Conversion Research Laboratory, ITB. Since 2008, he is a lecturer at School of Electrical Engineering and Informatics ITB, Indonesia. His research interests include power electronics and electrical machinery. Dr. Tri Desmana Rachmildha can be contacted trides@konversi.ee.itb.ac.id and trides@gmail.com



Yanuarsyah Haroen, He received B.Eng degree in electrical engineering from Institut Teknologi Bandung (ITB), Indonesia, in 1976. He received Diplôme Ingenieur, DEA/Master, and Docteur Ingenieur degrees from ENSEEIHT-INPT-France, in 1980, 1981, and 1983, respectively. He is a lectrurer in School of Electrical Engineering and Informatics since 1978 and became a Professor in 1999. His research interests include electrical machinery, power electronics and renewable energy.

He is also a senior member of IEEE, member of Professional Engineer in Indonesia and head of Indonesian Electric Machinery Society. Prof. Dr. Yanuarsyah Haroen can be contacted at yanuar@stei.itb.ac.id.



Pekik Argo Dahono, He got the Insinyur (Ir) degree, from the Institut Teknologi Bandung, Indonesia, in 1985, the Master and Doctor Engineering degrees from Tokyo Institute of Technology, Japan, in 1992 and 1995, respectively, all in electrical engineering.

He is registered as a Professional Engineer in Indonesia and ASEAN. He is a senior member of IEEE. He is cofounder of Indonesia Smart Grid Initiatives and Indonesia Power Quality Initiatives.

At present, he is a professor in electrical engineering at the Institut Teknologi Bandung. He has interest on power electronics, power systems, and power quality.